

Slice & Why it Matters

I chose to research Newick trees. It is one of the main techniques that Professor Novak suggested we use for data representation.

Key Findings

Newick trees are a form of phylogenetic tree – a tree diagram denoting the evolutionary relationship between various species – which are in a machine-readable format ([Allard, 2025](#)).

Newick trees have no official standard, but there was a formal grammar defined as part of the original usage of the Newick tree ([Czech, 2017](#)). Since denoting the phylogeny of species is a

monstrous task, it would be ideal to use an automated tool to encode phylogeny data into a

Newick tree for use with a Large Language Model. The Treemable program discussed in

“Treemable: A Graphical Tool to Generate Newick Strings from Phylogenetic Tree Images,” is a potential option for converting existing phylogenetic data into a Newick tree representation.

Since Newick trees are a string-based data structure, they consume a large quantity of resources.

As such, Python-based Newick tree modules struggle to handle trees with more than 20,000 leaves ([Junier, 2010](#)).

Implications for Our Team

- We will investigate using the Treemable program discussed by Allard to aid in the construction of Newick trees.
- We will consider using C when handling Newick strings because our data contains more than 40,000 species.

References

- Allard, J. B., & Kumar, S. (2025). Treemble: A Graphical Tool to Generate Newick Strings from Phylogenetic Tree Images. *arXiv preprint arXiv:2508.07081*.
- Czech, L., Huerta-Cepas, J., & Stamatakis, A. (2017). A critical review on the use of support values in tree viewers and bioinformatics toolkits. *Molecular biology and evolution*, 34(6), 1535-1542.
- Junier, T., & Zdobnov, E. M. (2010). The Newick utilities: high-throughput phylogenetic tree processing in the UNIX shell. *Bioinformatics*, 26(13), 1669-1670.